

- 6 (a) The *lac* operon is a section of DNA present in the genome of *Escherichia coli*. The structural genes of the *lac* operon are only fully expressed when the bacteria are exposed to high lactose concentrations.

Fig. 6.1 is a diagram showing the *lac* operon and a nearby region of the *E. coli* genome.

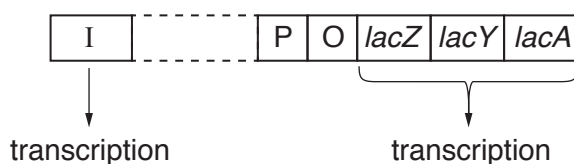


Fig. 6.1

- (i) Fig. 6.1 shows how the *lac* operon consists of structural genes and regulatory sequences.

Use Fig. 6.1 to identify two structural genes.

Complete Table 6.1 to name each structural gene and its product.

Table 6.1

structural gene	name of gene product

[2]

- (ii) Gene I is transcribed all the time to produce its protein. This is constitutive expression.

Explain why some genes show constitutive expression.

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 [1]

- (iii) Describe the effect of the product of gene I on the functioning of the *lac* operon.

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 [2]

(b) If *E. coli* is put into a nutrient medium containing lactose, some new enzymes are synthesised. These are described as inducible enzymes.

(i) Explain what is meant by an *inducible enzyme*.

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..... [2]

(ii) The structural genes of the *lac* operon are **not** expressed when lactose is absent.

Suggest **one** reason why this is beneficial to *E. coli*.

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..... [1]

[Total: 8]